

VINAY K R

Software Engineer | Applied AI Systems

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PROFESSIONAL SUMMARY

Software engineer with nearly three years of professional experience, including internship, across enterprise gaming and digital-asset custody. Built a foundation in typed product systems, APIs, authorization, caching, real-time workflows and testing; subsequently co-developed Medha's Go backend and system architecture. Building an Applied AI specialization through independent Python/FastAPI services, ML data applications and tested agent workflows.

TECHNICAL SKILLS

Applied AI & ML: Python, FastAPI, scikit-learn, PyTorch, LangGraph, agent workflows, SSE, OpenTelemetry
Services & data: Pydantic, TypeScript, Node.js, Go, REST APIs, SQL, PostgreSQL, PostGIS, Redis, caching
Quality & delivery: pytest, Mypy, Ruff, OpenTelemetry, Docker, GitHub Actions, Git, authentication, RBAC
Professional application stack: Angular, RxJS, C#.NET, SQL Server, WebSockets, Cypress

PROFESSIONAL EXPERIENCE

Software Engineer — Full Stack | Liminal Custody

Nov 2025 – Mar 2026 | Bengaluru

- Developed Angular custody-policy workflows and Node.js/TypeScript APIs with atomic bulk updates and transaction rollback handling.
- Implemented organization-scoped authorization and step-up authentication across Auth0, frontend flows and backend middleware; added Redis ownership and expiry checks for cached quotes.

Senior Associate Software Engineer | Light & Wonder

Aug 2023 – Jul 2025 | Bengaluru | Promoted from Associate Software Engineer

- Developed Angular/TypeScript workflows integrated with C#.NET APIs and SQL Server for financial and player-management applications.
- Fixed an RxJS/WebSocket subscription-lifecycle leak in gaming-device software; developed audit-data capture and reporting, maintained Cypress regression coverage, and connected crash reporting with backend telemetry.

Full-Stack Intern | Light & Wonder

Mar 2023 – Jul 2023

- Built an Angular and C#.NET game-recommendation application with feedback and ratings during a 16-week internship.

PROJECTS

FastAPI GenAI Agent Patterns | Personal reference implementation | Aug 2026

github.com/vi-nayKR/fastapi-genai-agent-patterns

- Built a typed LangGraph workflow with deterministic specialist routing, bounded iterations, in-memory checkpoint/resume, human approval and SSE events.
- Added real provider mode, structured outputs, hybrid retrieval (BM25 + dense semantic hash with RRF), and reproducible 30-case benchmark evaluation; 49 tests passed, Ruff and strict Mypy.

Medha | Co-developed product — backend and systems contribution | 2026

github.com/vi-nayKR/medha-platform-api

- Co-developed Go APIs using PostgreSQL/PostGIS, Redis and WebSockets, with layered domain, service and repository boundaries; implemented transaction-aware database access and contributed to authentication, location-based discovery and runtime architecture.

PUBLICATIONS

Data Visualisation of Time-Tradable Assets Using Machine Learning | IEEE Xplore | 2023

ieeexplore.ieee.org/document/10275962

- Co-authored IEEE conference publication (DOI: 10.1109/NCNSP56992.2023.10275962) evaluating ML models (Linear Regression, Decision Tree, SVR, LSTM) and error metrics on financial time-series.
- Engineered accompanying interactive market-data application using FastAPI and Angular to visualize OHLCV indicators, compare model predictions, and demonstrate time-series analytics.

EDUCATION

B.E., Computer Science & Engineering | Siddaganga Institute of Technology | 2023 | CGPA: 8.65/10